

Does the clustering keep “fecundity” together — and does silhouette know?

Tracking **72 fecundity traits** (offspring-quantity concept) through HDBSCAN clustering as fields are added V1→V8, with V20 as an all-identity-fields control. Each cell is the cluster a trait landed in; **same colour within a column = same cluster**. The punchline: the version with the **highest silhouette groups fecundity the worst**.

HIGHEST SILHOUETTE — V1 (NAME ONLY)

0.72

...yet fecundity is split across **4 clusters** with 19 in noise. Only **33%** share a cluster. The metric is hijacked by tight name-only blocks.

BEST GROUPING — V6 (+ DEFINITION, UNIT, TAXONOMY, STAGE)

94%

of fecundity traits land in **one** cluster (only 4 in noise) — but silhouette here is just **0.29**. Correct grouping ≠ high silhouette.

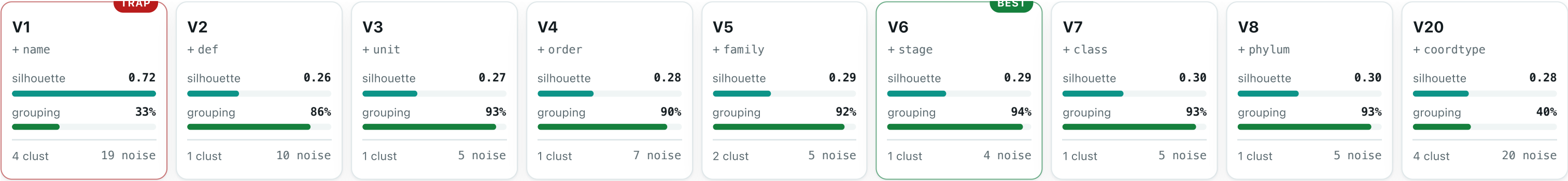
CONTROL — V20 (ALL IDENTITY FIELDS ADDED)

4x

clusters again (20 in noise, 40% dominant). Genus/species/lat-long re-scatter fecundity by organism & place — the identity-field hijack.

Per-version: silhouette vs. how well fecundity is grouped

Two bars per version. **Silhouette** is the global metric being optimised; **grouping** is the share of fecundity traits sitting in one dominant cluster (our real goal). Watch them cross: silhouette is highest at V1 exactly where grouping is worst.



The full tracking matrix 72 traits × 9 versions

A solid block of one colour down a column means fecundity is being clustered together. Hatched cells are noise (unclustered, −1). Rows sorted by trait name; definitions distinguish the individual records.

Trait name	Definition	V1 +name	V2 +def	V3 +unit	V4 +order	V5 +family	V6 +stage	V7 +class	V8 +phylum	V20 +coordtype
Oviposition Rate	mean number of eggs per female	—	—	—	—	—	—	—	—	—
Reproductive average daily fecundity	the number of newborn nymphs (N1) per female per day	—	10	4	4	5	3	3	3	—
Reproductive average total fecundity	the number of newborn nymphs (N1) per female	—	10	4	4	5	3	3	3	—
age specific fecundity	mean number of female offspring individual-1 at specific age	—	—	4	—	5	3	3	3	—
age-specific daily fecundity	new nymphs produced each day	—	10	4	4	5	3	3	3	—
average daily fecundity	daily fecundity of female aphids	—	—	4	—	5	3	3	3	—
effective fecundity	mean number of nymphs produced individual-1 during a period equivalent to the prereproductive time	—	10	4	4	5	3	3	3	12
egg count	mean number of eggs layed	10	—	—	—	6	—	—	—	—
fecundity	individual-level lifetime eggs	30	10	4	4	5	3	3	3	10
fecundity	individual-level mean eggs per gonotrophic cycle	30	10	4	4	5	3	3	3	10
fecundity	individual-level weight of lifetime eggs per individual	30	10	4	4	5	3	3	3	14
fecundity	mean duration of life stage	30	10	4	4	5	3	3	3	12
fecundity	mean eggs individual-1	30	10	4	4	5	3	3	3	—
fecundity	mean eggs produced per plant	30	10	4	4	5	3	3	3	13
fecundity	mean lifetime egg punctures	30	10	4	4	5	3	3	3	—
fecundity	mean lifetime eggs	30	10	4	4	5	3	3	3	10
fecundity	mean lifetime eggs individual-1	30	10	4	4	5	3	3	3	13
fecundity	mean lifetime eggs laid per female	31	10	4	4	5	3	3	3	14
fecundity	mean lifetime eggs per 10 individuals	30	10	4	4	5	3	3	3	10
fecundity	mean lifetime eggs per individual	30	10	4	4	5	3	3	3	13
fecundity	mean lifetime eggs per individual (across all humdity levels)	30	10	4	4	5	3	3	3	13
fecundity	mean lifetime eggs produced by an individual female	30	10	4	4	5	3	3	3	14
fecundity	mean lifetime nymphs individual-1	31	10	4	4	5	3	3	3	12
fecundity	mean lifetime offspring	30	10	4	4	5	3	3	3	12
fecundity	mean lifetime offspring as a function of size and adult and growth temperature	30	10	4	4	5	3	3	3	13
fecundity	mean lifetime progeny	30	10	4	4	5	3	3	3	12
fecundity	mean lifetime progeny per female	30	10	4	4	5	3	3	3	12
fecundity	mean lifetime total number of nymphs	31	10	4	4	5	3	3	3	12
fecundity	mean liftetime eggs	31	10	4	4	5	3	3	3	—
fecundity	mean number of eggs per cycle	30	10	4	4	5	3	3	3	10
fecundity	mean number of eggs per cycle (1st)	31	10	4	4	5	3	3	3	10
fecundity	mean number of female offspring individual-1	30	10	4	4	5	3	3	3	—
fecundity	mean number of lifetime eggs by a tick	31	10	4	4	5	3	3	3	14
fecundity	mean number of progeny in a reproductive period equal to development time	30	10	4	4	5	3	3	3	12

Trait name	Definition	V1 +name	V2 +def	V3 +unit	V4 +order	V5 +family	V6 +stage	V7 +class	V8 +phylum	V20 +coordtype
fecundity	mean of daily newborn nymph counts	30	10	4	4	5	3	3	3	12
fecundity	mean offspring per reproduction day	30	10	4	4	5	3	3	3	12
fecundity	mean population-level offspring (100 individuals)	31	10	4	4	5	3	3	3	12
fecundity	number of aphids produced	30	10	4	4	5	3	3	3	12
fecundity	total number of eggs produced	30	10	4	4	5	3	3	3	14
fecundity	total population-level number of offspring produced	31	10	4	4	5	3	3	3	12
fecundity (alate)	mean lifetime offspring	—	10	4	4	5	3	3	3	12
fecundity (apterous)	mean lifetime offspring	—	10	4	4	5	3	3	3	12
fecundity rate	maximum number of eggs produced in a single day by an individual female	29	10	4	4	5	3	3	3	14
fecundity rate	maximum number of offspring per female per day	29	10	4	4	5	3	3	3	12
fecundity rate	maximum progengy individual-1 day-1	29	10	4	4	5	3	3	3	12
fecundity rate	maximum progeny per unit time	29	10	4	4	5	3	3	3	12
fecundity rate	mean duration of life stage	29	10	4	4	5	3	3	3	12
fecundity rate	mean eggs individual-1 day-1	29	10	4	4	5	3	3	3	—
fecundity rate	mean eggs per unit time	29	10	4	4	5	3	3	3	13
fecundity rate	mean eggs produced by an individual female per day	29	10	4	4	5	3	3	3	14
fecundity rate	mean lifetime offspring	29	10	4	4	5	3	3	3	12
fecundity rate	mean maximum progeny per unit time	29	10	4	4	5	3	3	3	12
fecundity rate	mean number of eggs individual-1 day-1	29	10	4	4	5	3	3	3	13
fecundity rate	mean number of nymphs per day per female	29	10	4	4	5	3	3	3	12
fecundity rate	mean number of nymphs produced individual-1 during a period equivalent to the prereproductive time	29	10	4	4	5	3	3	3	12
fecundity rate	mean number of offspring per reproductive cycle	29	10	4	4	5	3	3	3	—
fecundity rate	mean offspring per day	29	10	4	4	5	3	3	3	13
fecundity rate	mean offspring per reproduction day	29	10	4	4	5	3	3	3	12
fecundity rate	mean offspring per unit time	29	10	4	4	5	3	3	3	12
fecundity rate	mean progeny per unit time	29	10	4	4	5	3	3	3	12
fecundity rate	mean weight of egg mass per day	29	10	4	4	5	3	3	3	14
fecundity rate	peak number of eggs per day per individual	29	10	4	4	5	3	3	3	14
fecundity rate (alate)	mean offspring per unit time	—	10	4	4	5	3	3	3	12
fecundity rate (apterous)	mean offspring per unit time	—	10	4	4	5	3	3	3	12
maximum fecundity	mean duration from onset of reproduction to maximum reproductive output	—	—	4	4	5	3	3	3	—
mean fecundity	mean number of offspring produced by females	—	10	4	4	5	3	3	3	12
mean fecundity	number of nymphs deposited	—	10	4	4	5	3	3	3	—
peak fecundity	day post engorgement on which an individual female laid most eggs	—	—	—	—	—	3	3	3	—

Trait name	Definition	V1 +name	V2 +def	V3 +unit	V4 +order	V5 +family	V6 +stage	V7 +class	V8 +phylum	V20 +coordtype
reproductive capacity	mean lifetime nymphs (30 days)	—	—	4	4	—	3	—	—	—
reproductive capacity	proportion of individuals that laid eggs after first blood meal	—	—	—	—	—	—	—	—	—
fertility ♦	number of offspring produced	—	—	4	4	5	3	3	3	—
nymph production ♦	number of nymphs produced / female	—	—	—	—	—	—	—	—	—

 Noise (−1, unclustered)  Dominant cluster in that column  Included via definition (name isn’t a fecundity term)

How to read this.

- **V1 → V2 is the key jump.** Name-only (V1) scatters fecundity into 3–4 tight name-string clusters (29/30/31) that inflate silhouette to 0.72. Adding the **definition** (V2) collapses ~94% of them into a single semantic cluster while silhouette *drops* — the metric and the goal point in opposite directions.
- **V3–V8 stay stable and correct.** Adding unit, then the identity fields order/family/stage/class/phylum barely moves fecundity: it stays in one dominant cluster. So at this depth the identity fields don’t yet hijack — the trait signal still dominates.
- **V20 breaks it.** Once genus, species, habitat, location and lat/long are in the text, fecundity re-fragments across 4 clusters (colours 10/12/13/14) grouped by organism and locality, not by trait — with silhouette essentially unchanged (0.28). This is the homogeneous-identity-field hijack, made visible.
- **Curation errors ride along.** Rows named *fecundity* but defined as “mean duration of life stage” still track the true fecundity cluster through V2–V8 — the name text carries them in, so mislabelled records are grouped *as if* fecundity.
- **♦ definition-included rows.** *fertility* ("number of offspring produced") joins the fecundity cluster from V3 on; *nymph production* stays noise throughout.

Bottom line. Silhouette cannot be trusted to pick the best version for searchability: it rewards V1 (worst) and is flat across V6 (best) and V20 (hijacked). Grouping share against a fecundity ground-truth separates them cleanly.